POLITECNICO MILANO 1863

BASI DI DATI 1
Academic Year 2024/2025
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Exam January 20th, 2025 **DURATION: 2 hours**

Part 1 – QUERY LANGUAGES

Exercise 1 – SQL (13pt)

Consider the following database of a company that stores the attendance of its employees in some training courses.
The `ExamPassed` column indicates whether the exam scheduled at the end of the course is passed or not.
There are no courses that last more than one day.
Note that the database also tracks the time employees actually participate in the course: this value may be lower than the nominal duration of the course.
There are no participations with `ParticipationTime` equal to zero.

EMPLOYEE(BadgeNo, FirstName, LastName, HiringDate, City)
COURSE(Code, Date, Title, Topic, Duration)
ATTENDING(BadgeNo, CourseCode, CourseDate, ParticipationTime, ExamPassed)

A. (2pt) Define the schema of the ATTENDING table, including all the necessary constraints.
B. (3pt) Find the employees who passed the exams of at least 3 courses they have not attended entirely.
C. (4pt) Find the topic with the highest number of courses where no employees participated.
D. (4pt) Describe clearly and concisely the result of the following query:

```
CREATE VIEW aview(Topic, BadgeNo, Counter) AS
    SELECT Topic, BadgeNo, COUNT(*)
    FROM Attending, Course
    WHERE CourseCode = Code
            AND CourseDate = Date
            AND ExamPassed = True
    GROUP BY Topic, BadgeNo
```

```
SELECT BadgeNo, COUNT(*)
FROM aview AS x
WHERE Counter >= ALL (
    SELECT y.Counter
    FROM aview AS y
    WHERE x.Topic = y.Topic
)
GROUP BY BadgeNo
```

Exercise 2 – Formal languages (6pt)

Write the following query in Relational Algebra (3pt) and Datalog (3pt):

Find the employees who have attended and passed all the available courses.

PART 2 – DATABASE DESIGN

Exercise 3 – Theory (1pt)

Define the concept of integrity constraint: describe the main categories and give some examples.

Exercise 4 – Conceptual and logical design (13pt)

- *The project evaluation is based on its completeness, correctness, legibility, minimality and clarity.*
- *Remember to specify the **identifiers** of all the entities and the **cardinality** of all the associations.*

A theme park offers a variety of attractions that customers can visit by purchasing tickets and passes either online through the digital platform or directly on-site. For online purchases, customers register with their first name, last name, email address, and phone number. Each customer (online or on-site) can purchase different types of tickets: daily tickets or passes (seasonal, annual, or customized for a specific period). Each ticket is associated with a validity period and has a price.

The park is organized into various attractions, each with a unique name, description, category (e.g., roller coaster, carousel, Ferris wheel), opening and closing times, and a maximum visitor capacity per cycle. Some attractions may have specific age group requirements or access restrictions (e.g., a minimum height). The attractions require regular maintenance to ensure safety and optimal operation. Each maintenance task is recorded with the start date, expected duration, type of intervention (e.g., "Inspection," "Repair," "Upgrade"), and a final report summarizing the work performed. During the maintenance period, the availability status of the attraction is updated in real time (e.g., "Unavailable," "Reduced Operation"). The system maintains a complete history of all maintenance tasks performed for each attraction.

For every visit to the park, the entry time and the attractions visited are recorded. Daily tickets allow a single entry to the park, while subscriptions permit multiple entries, each of which is tracked separately. The park can organize special events such as evening shows or themed parties. Each event has a name, description, date, and time, and may require a specific ticket or be included in certain types of passes.